

ESP32 Real LED

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Abstract

This activity aims to introduce and implement the control of a simple electronic device, namely an LED, directly using the ESP32 microcontroller. The implementation was carried out in real-time during class as part of a hands-on learning session in microcontroller and Internet of Things (IoT) applications. Students were guided to understand the basics of using the ESP32, including connecting an LED to a digital pin and writing control code using Visual Studio Code with the PlatformIO extension. The LED was successfully controlled with immediate response based on the uploaded program. This project demonstrates that the ESP32, combined with modern development tools like VS Code, offers an efficient and user-friendly platform for learning basic digital device control. Conducting the session live in class provided students with valuable practical experience and significantly improved their understanding of embedded systems programming.

Keywords—Internet of Things, ESP32, LED.

1. Introduction

1.1 Background

The rapid development of microcontroller technology has made it easier for students and hobbyists to explore embedded systems and Internet of Things (IoT) applications. One of the widely used microcontrollers today is the ESP32, known for its built-in Wi-Fi and Bluetooth capabilities, multiple GPIO pins, and compatibility with various development environments.

Learning to control basic electronic components such as LEDs is a fundamental step in understanding how microcontrollers interact with hardware. Through this activity, students are introduced to real-time embedded system programming using the ESP32 and Visual Studio Code (VS Code) as the development environment. VS Code, combined with the PlatformIO extension, provides an efficient and modern workflow for writing, uploading, and debugging code.

This practical session was conducted in-class to allow students to directly observe and experience how the code they write can affect hardware behavior in real time. The activity not only helps students strengthen their programming logic, but also fosters a deeper understanding of hardware-software interaction, which is essential for developing more complex IoT systems in the future.

1.2 Objectives

- To understand the basic use of the ESP32 microcontroller, particularly in controlling simple electronic components such as an LED.
- To implement microcontroller programming using Visual Studio Code and the PlatformIO extension as the development environment.
- To train students in connecting hardware with software through hands-on practice in the classroom.
- To enhance students' understanding of basic concepts in embedded systems and the Internet of Things (IoT) through real-time experiments.
- To encourage active and participatory learning, where students can immediately observe the effects of their code on physical devices.

2. Methodology

2.1 Tools & Materials

Hardware:

- ESP32 development board
- Breadboard
- LED (any color)
- Jumper wires
- Laptop with Visual Studio Code and PlatformIO installed

Software:

- Visual Studio Code with the PlatformIO extension
- A new PlatformIO project was created in Visual Studio Code.
- Required libraries were added via `platformio.ini`.

2.2 Implementation Steps

Project Creation:

A. Create a New Project

1. Click the PlatformIO Home (alien icon) on the sidebar.
2. Click "New Project".
3. Fill in:
 - Project Name: e.g., `esp23_real_led`
 - Board: Select DOIT ESP23 DEVKIT V1
 - Framework: Arduino
 - Location: default location
4. Click Finish — PlatformIO will create a project with the right structure.

Physical Connection:

- Connect ESP32 to the sensors and actuators using jumper wires.
- Ensure proper connections for the sensor's power, ground, and signal pins.
- Use a breadboard to organize the components and avoid messy connections.

Code Development:

```
#include <Arduino.h> // Wajib untuk PlatformIO + ESP32

// Deklarasi pin LED
int lampu = 33;
int lampu2 = 26;

void setup() {
    Serial.begin(115200); // Inisialisasi komunikasi Serial
    Serial.println("ESP32 Blinking LED");

    // Atur pin sebagai OUTPUT
    pinMode(lampu, OUTPUT);
    pinMode(lampu2, OUTPUT);
}

void loop() {
    // Nyalakan kedua LED
    digitalWrite(lampu, HIGH);
    Serial.println("LED ON");

    delay(1000); // Tunggu 1 detik

    // Matikan kedua LED
    digitalWrite(lampu, LOW);
    Serial.println("LED OFF");

    delay(500); // Tunggu 1 detik sebelum mengulang

    // Nyalakan kedua LED
```

```

void loop() {
    // Nyalakan kedua LED
    digitalWrite(lampu, HIGH);
    Serial.println("LED ON");

    delay(1000); // Tunggu 1 detik

    // Matikan kedua LED
    digitalWrite(lampu, LOW);
    Serial.println("LED OFF");

    delay(500); // Tunggu 1 detik sebelum mengulang

    // Nyalakan kedua LED
    digitalWrite(lampu2, HIGH);
    Serial.println("LED ON");

    delay(500); // Tunggu 1 detik

    // Matikan kedua LED
    digitalWrite(lampu2, LOW);
    Serial.println("LED OFF");

    delay(500); // Tunggu 1 detik sebelum mengulang
}

```

Simulation & Verification:

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3. Results and Discussion



